21. A homogeneous differential equation of the form $\frac{dy}{dx} = f\left(\frac{y}{x}\right)$ can be solved by making substitution,

(a) $y = v\,x$ (b) $v = y\,x$

(c) $x = v\,y$ (d) $x = v$

22. A homogeneous differential equation of the form $\frac{dx}{dy} = f\left(\frac{x}{y}\right)$ can be solved by making substitution,

(a) $x = v\,y$ (b) $y = v\,x$

(c) $y = v$ (d) $x = v$

23. The variable separable form of $\frac{dy}{dx} = \frac{y(x-y)}{x(x+y)}$ by taking $y = vx$ and $\frac{dy}{dx} = v + x\frac{dv}{dx}$ is

(a) $\frac{2v^2}{1+v}dv = \frac{dx}{x}$

(b) $\frac{2v^2}{1+v}dv = -\frac{dx}{x}$

(c) $\frac{2v^2}{1-v}dv = \frac{dx}{x}$

(d) $\frac{1+v}{2v^2}dv = -\frac{dx}{x}$

24. Which of the following is the homogeneous differential equation?

(a) $(3x-5)\,dx = (4y-1)\,dy$

(b) $xy\,dx - (x^3+y^3)\,dy = 0$

(c) $y^2\,dx + (x^2 - xy - y^2)\,dy = 0$

(d) $(x^2+y)\,dx = (y^2+x)\,dy$

25. The solution of the differential equation $\frac{dy}{dx} = \frac{y}{x} + \frac{f\left(\frac{y}{x}\right)}{f'\left(\frac{y}{x}\right)}$ is

(a) $f\left(\frac{y}{x}\right) = k\,x$ (b) $x\,f\left(\frac{y}{x}\right) = k$

(c) $f\left(\frac{y}{x}\right) = k\,y$ (d) $y\,f\left(\frac{y}{x}\right) = k$

Miscellaneous Problems

1. Suppose that $Q_d = 30 - 5P + 2\frac{dP}{dt} + \frac{d^2P}{dt^2}$ and $Q_s = 6 + 3P$. Find the equilibrium price for market clearance.

2. Form the differential equation having for its general solution $y = ax^2 + bx$

3. Solve $yx^2dx + e^{-x}dy = 0$

4. Solve $(x^2 + y^2)dx + 2xy\,dy = 0$

5. Solve $x\frac{dy}{dx} + 2y = x^4$

6. A manufacturing company has found that the cost C of operating and maintaining the equipment is related to the length 'm' of intervals between overhauls by the equation $m^2\frac{dC}{dm} + 2mC = 2$ and c = 4 and when m = 2. Find the relationship between C and m.

7. Solve $(D^2 - 3D + 2)y = e^{4x}$ given y = 0 when x = 0 and x = 1.

8. Solve $\frac{dy}{dx} + y\cos x + x = 2\cos x$

9. Solve $x^2ydx - (x^3 + y^3)dy = 0$

10. Solve $\frac{dy}{dx} = xy + x + y + 1$

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